

JUnit/Mockito Exercise 6: Mockito Mock Dependencies Repository

Mocking Dependencies in Spring Tests

using Mockito-

Exercise 1: Mocking a Service Dependency in a Controller Test

Task: Write a unit test for a Spring controller that uses a service to fetch data. Mock the service dependency using Mockito.

Step-by-Step Solution:

1. **Create the User Entity:**

@Entity

```
public class User {
```

```
    @Id
```

```
    private Long id;
```

```
    private String name;
```

```
    // getters and setters
```

```
}
```

2. **Create the UserService:**

@Service

```
public class UserService {
```

```
    @Autowired
```

```
    private UserRepository userRepository;
```

```
    public User getUserById(Long id) {
```

```
        return userRepository.findById(id).orElse(null);
```

```
    }
```

```
}
```

3. **Create the UserController:**

@RestController

@RequestMapping("/users")

```
public class UserController {
```

```
    @Autowired
```

```
    private UserService userService;
```

```
    @GetMapping("/{id}")
```

```
    public ResponseEntity<User> getUser(@PathVariable Long id) {
```

```
        return ResponseEntity.ok(userService.getUserById(id));
```

```
    }
```

```
}
```

4. **Create the UserControllerTest:**

Write code for this.

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Exercise 2: Mocking a Repository in a Service Test

****Task:**** Write a unit test for a Spring service that uses a repository to fetch data. Mock the repository dependency using Mockito.

Step-by-Step Solution:

1. ****Create the User Entity:****

@Entity

```
public class User {  
    @Id  
    private Long id;  
    private String name;  
    // getters and setters  
}
```

2. ****Create the UserRepository:****

```
public interface UserRepository extends JpaRepository<User, Long> {  
}
```

3. ****Create the UserService:****

@Service

```
public class UserService {  
    @Autowired  
    private UserRepository userRepository;  
    public User getUserById(Long id) {  
        return userRepository.findById(id).orElse(null);  
    }  
}
```

4. ****Create the UserServiceTest:****

Write code for this.

Exercise 3: Mocking a Service Dependency in an Integration Test

****Task:**** Write an integration test for a Spring Boot application that mocks a service dependency using Mockito.

Step-by-Step Solution:

1. ****Create the User Entity:****

@Entity

```
public class User {  
    @Id  
    private Long id;  
    private String name;
```

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```
// getters and setters
}

2. **Create the UserService:**

@Service
public class UserService {

    @Autowired
    private UserRepository userRepository;

    public User getUserById(Long id) {
        return userRepository.findById(id).orElse(null);
    }
}

3. **Create the UserController:**

@RestController
@RequestMapping("/users")
public class UserController {

    @Autowired
    private UserService userService;

    @GetMapping("/{id}")
    public ResponseEntity<User> getUser(@PathVariable Long id) {
        return ResponseEntity.ok(userService.getUserById(id));
    }
}

4. **Create the UserIntegrationTest:**

Write code for this.

Hint: Use '@SpringBootTest', '@AutoConfigureMockMvc'
```